7 **(a)** For a steady and streamline flow of an incompressible and non-viscous fluid, the total hydraulic head H of the flow through a particular cross-section of a pipe is given by the expression

$$H = \frac{P}{\rho g} + \frac{v^2}{2g} + h$$

where P is the pressure of the fluid at the cross-section,
ρ is the density of the fluid,
v is the velocity of the fluid at the cross-section, and
h is the height of the centreline of pipe above a reference level.

H is related the energy in the fluid.

Suggest the type of energy associated with the following terms:

(i) $\frac{v^2}{2g}$

..... [1]

(ii) h

..... [1]

(b) Fig. 7.1 shows a steady and fully developed flow through a horizontal pipe of varying diameter. The inlet has an inner pipe diameter D_A and fluid velocity V_A while the outlet has an inner pipe diameter D_B and fluid velocity V_B.

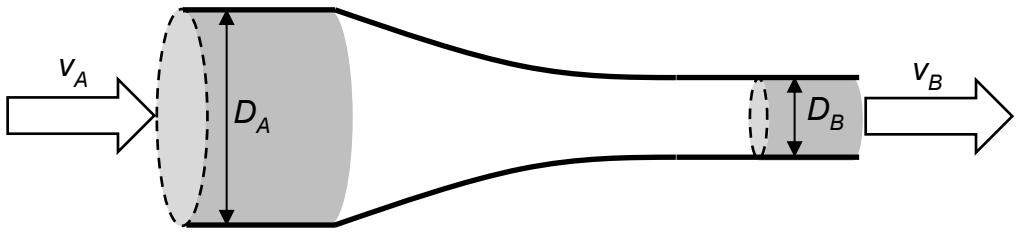

Fig. 7.1

(i) Assuming there are no energy losses within the pipe, state how the total hydraulic head at the inlet and at the outlet are related.

..... [1]

(ii) If the fluid can be considered incompressible, show that v_A and v_B are related by the expression

$$\frac{v_A}{v_B} = \frac{D_B^{\ 2}}{D_A^{\ 2}}$$ [2]

(iii) The inner diameter of the inlet is 0.100 m and the inner diameter of the outlet is 0.050 m. Water flows into the inlet at a speed of 8.0 m s^{-1}.

Using the hydraulic head expression given in **(a)** and your answers to **(b)**, determine the pressure difference between the inlet and outlet. (Density of water is 1000 kg m^{-3}.)

pressure difference = …… Pa [3]

(c) In a typical flow between 2 cross-sections of a pipe, there will be energy losses due to frictional forces between the walls of the pipe and the fluid. This is accounted for by including an additional head loss term,

$$H_{Loss} = f_D \frac{v^2 L}{2gD}$$

where f_D is the friction factor,
L is the length of the pipe, and
D is the inner diameter of the pipe.

(i) State how this additional term will affect your answer to **(b)(iii)**.

.....

..... [1]

(ii) If the pipe in **(b)** has a length of 10 m and a friction factor of 0.0038, estimate the magnitude of the difference to your answer in **(b)(iii)** if frictional losses are considered.

difference = ... Pa [2]

(d) A student suggests using water flow in rigid pipes to model blood flow in blood vessels.

(i) State a difference between water flow in rigid pipes and blood flow in blood vessels.

.....

.....

..... [1]

(ii) The accumulation of fat and cholesterol deposits along the arterial walls has an effect of narrowing the arteries leading to an increase in blood pressure.

State and explain if the modelling proposed by the student will able to correctly predict this observation.

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..... [3]